# Predicting Patient Readmission Risk using Machine Learning

**Team:** Saumya Padhi, Surendar Reddy, Vignesh Nagarajan

**Collaboration:** https://github.com/saumyasam/Hospial_Readmission_ML.git

MGT-665-NW Solv Probs W/ Machine Learning Graduate

DeVos Graduate School, Northwood University

**Dr. Mighty Itauma Itauma**

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## Problem Statement

Unplanned hospital readmissions are costly, both financially and in terms of patient health outcomes. Identifying patients at higher risk of readmission using machine learning can enable healthcare providers to take proactive, personalized action. This project explores predictive and unsupervised learning methods to classify patients based on their likelihood of readmission and cluster them by wellness profiles for targeted care interventions.

## Issue Being Solved

The central question this project addresses is: Can machine learning accurately predict hospital readmission and identify patient clusters that indicate wellness profiles? The goal is to combine classification and clustering techniques to support targeted healthcare interventions and improve hospital resource planning.

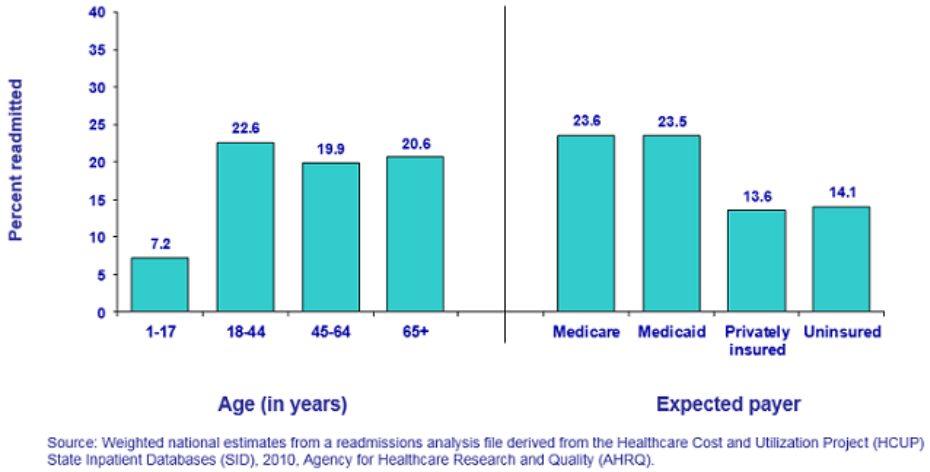
## Introduction

Diabetes mellitus is among the top causes of hospital readmissions globally. Its chronic nature requires ongoing monitoring and often results in recurrent hospitalization, the majority of which are within 30 days of discharge. Readmission prediction is paramount to prevention and relief of health infrastructure overload. Previous studies have shown that predictive analytics through machine learning application can assist in the forecasting of hospital outcomes such as readmission, mortality, and length of stay (Rajkomar, Dean, & Kohane, 2019). Yet, constraints such as large data feature sets, poor data quality, and class imbalances limit the performance of traditional models. Predictive performance is

enhanced in this research work by utilizing various machine learning models over a large diabetes dataset to classify high-risk patients before readmission.

**Figure 1**

**All-cause 30-day readmission rates for patients hospitalized with diabetes and complications by age and payer, U.S. hospitals (2010)**




*Note: This is a bar chart showing percent readmitted by age in years and by expected payer. Ages 1 through 17, 7.2 percent. Ages 18 through 44, 22.6 percent. Ages 45 through 64, 19.9 percent. Ages 65 and older, 20.6 percent. Medicare payer, 23.6 percent. Medicaid payer, 23.5 percent. Privately insured, 13.6 percent. Uninsured, 14.1 percent. Source: Weighted national estimates from a readmissions analysis file derived from the Healthcare Cost and Utilization Project State Inpatient Databases, 2010, Agency for Healthcare Research and Quality.*

**Abstract**

Reducing hospital readmissions is a critical concern for healthcare systems aiming to improve patient outcomes and lower costs. This study leverages machine learning to

predict hospital readmission risk and uncover patient wellness profiles through clustering. Using the UCI Diabetes Hospital Readmission dataset, which includes over 100,000 hospital visits, we examine various clinical and administrative variables such as number of prior inpatient visits, medication changes, and primary diagnoses. The project applies supervised classification techniques—K-Nearest Neighbors (KNN), Decision Trees, Naive Bayes, Random Forests, and Neural Networks—to predict readmission. Unsupervised clustering using KMeans and Agglomerative Clustering, coupled with PCA visualization, is also used to explore patient segmentation. The aim is twofold: to develop accurate predictive models and to identify actionable subgroups of patients for care optimization. This interim report presents preprocessing, model development, and initial evaluation metrics to establish a baseline for further refinement.

**Related Work**

Healthcare analytics has increasingly used machine learning in recent work to address challenging prediction issues. Rajkummar et al. (2019) also canvassed a broad range of machine learning techniques on electronic health records, referencing the capability of deep learning to handle sparse high-dimensional data. Zhou, Pan, Wang, and Vasilakos (2020) wrote on the potential of big data to aid healthcare systems and promoted the application of predictive modeling in making decisions. Despite these advances, previous models have struggled with class imbalance, particularly in the instance of predicting infrequent events such as early readmission. This work builds on such studies by contrasting and comparing the performance of standard classifiers and neural networks on

an extremely imbalanced data set and exploring both clustering and feature engineering methods for model performance improvement.

**Methodology**

The dataset used for the analysis was acquired from the UCI Machine Learning Repository and has over 100,000 diabetic hospitalization data. Preprocessing involved handling missing values, dropping columns with over 50% missing values, encoding categorical attributes using label encoding for nominal and one-hot encoding for binary features. The target, hospital readmission in 30 days, was handled as a binary variable via encoding. Clustering models such as KMeans and Agglomerative Clustering were employed within patient segmentation, with visualization being carried out through Principal Component Analysis (PCA).

**Data and Variables**

The data comes from the UCI Machine Learning Repository: the Diabetes 130-US hospitals readmission dataset. It contains 101,766 records across 10 years. Key variables include:

- **Demographics**: Age, race, gender
- **Administrative**: Admission type, discharge disposition, time in hospital
- **Clinical**: Number of procedures, medications, diagnoses, prior inpatient visits
- **Target**: Readmission within 30 days (Yes/No)

**Algorithms and Methods**

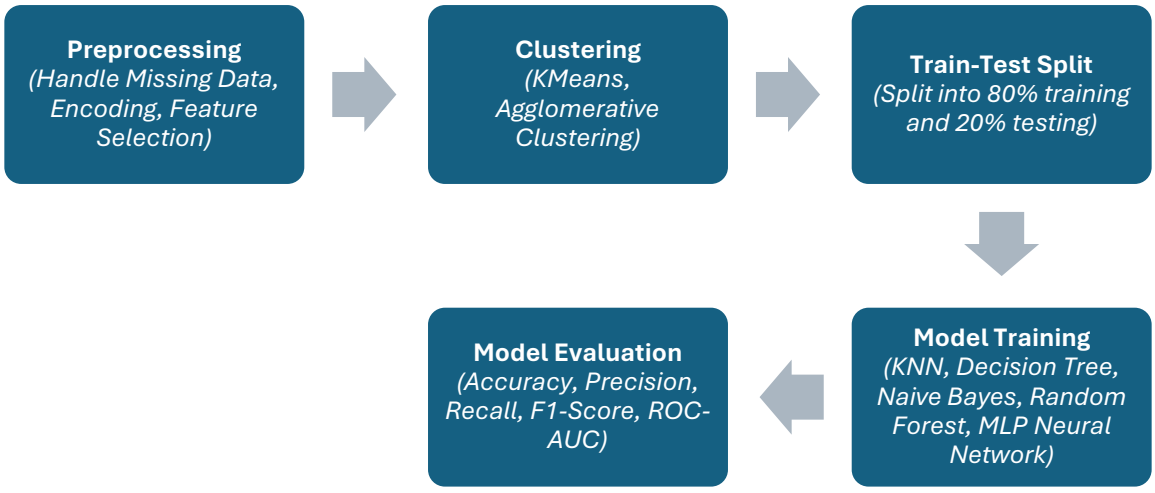
The following machine learning techniques are being applied:

- **Classification**: K-Nearest Neighbors (KNN), Decision Tree, Naive Bayes, Random Forest, and Multi-layer Perceptron (Neural Network)
- **Clustering**: KMeans and Agglomerative Clustering
- **Dimensionality Reduction**: Principal Component Analysis (PCA)
- **Evaluation**: Accuracy, F1-score, and classification reports

The models were evaluated based on accuracy, precision, recall, F1-score, and ROC-AUC using an 80/20 train-test split. Standard scaling was applied to numerical features before model training.

**Figure 2**

**Machine Learning Pipeline for Readmission Prediction**

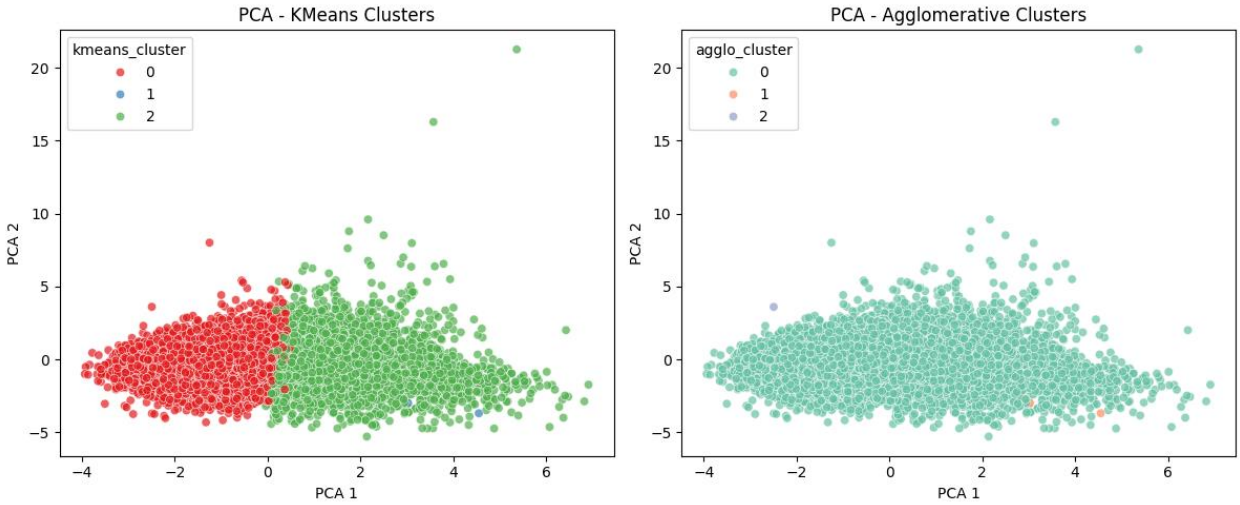



*Note: A visual representation of the machine learning workflow used in this study, beginning with data preprocessing, followed by unsupervised clustering, model training using five classifiers,*

*and evaluation using standard metrics.*

**Figure 3**

**Clustering Using PCA – KMeans and Agglomerative Comparison**




*Note: Two-dimensional PCA plots showing patient clusters formed by KMeans and Agglomerative algorithms for pattern discovery.*

**Results**

As described in Table 1, Accuracy level varied in models. Random Forest was best at 88.8%, narrowly followed by KNN at 88.0%. MLP neural network was at 83.7%, and Decision Tree was at 82.3%. Naive Bayes was lowest at 15.0% and indicates its vulnerability to data distribution and feature independence assumptions. F1-score and ROC-AUC performance metrics also indicated the robustness of ensemble and KNN models. Random Forest model both achieved highest weighted F1-score and ROC-AUC, the optimal trade-off between precision and recall. All models fared poorly on the minority class, 30-day readmitted patients. This is to be expected in the issue of sparse event prediction in clinical datasets.

**Table 1**

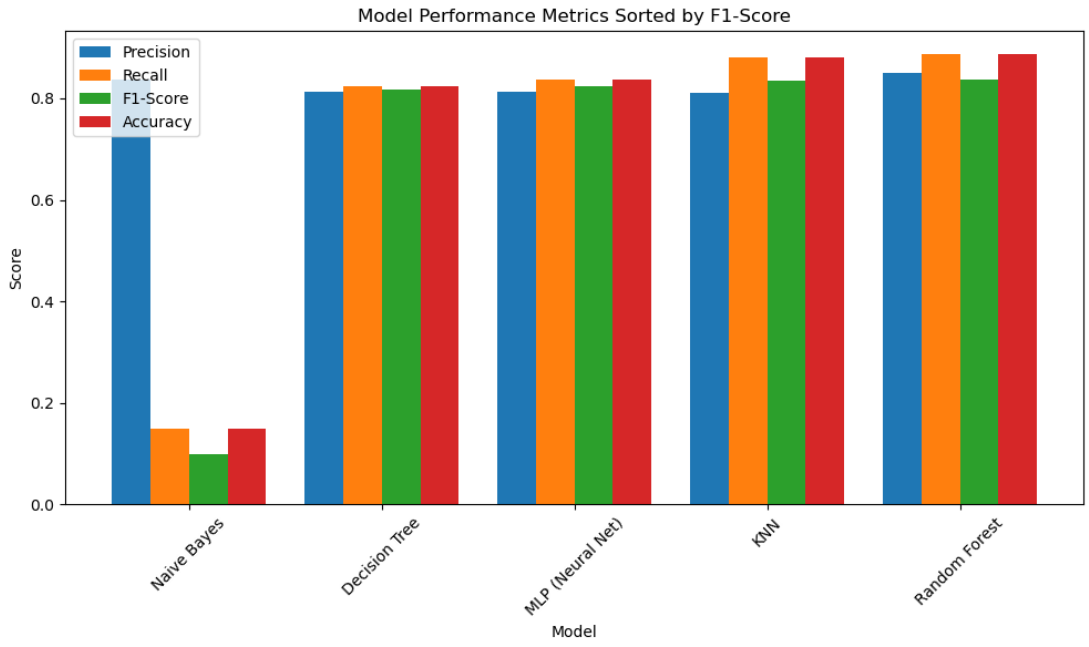
**Model Evaluation Metrics Summary**

| MODEL | PRECISION | RECALL | F1-SCORE | ACCURACY |
|---|---|---|---|---|
| **KNN** | 0.810 | 0.880 | 0.835 | 0.880 |
| **DECISION TREE** | 0.814 | 0.823 | 0.818 | 0.823 |
| **NAIVE BAYES** | 0.837 | 0.150 | 0.100 | 0.150 |
| **RANDOM FOREST** | 0.850 | 0.888 | 0.836 | 0.888 |
| **MLP (NEURAL NET)** | 0.812 | 0.837 | 0.824 | 0.837 |

*Note: Table with accuracy, precision, recall, F1-score — as shown earlier*

**Figure 4**

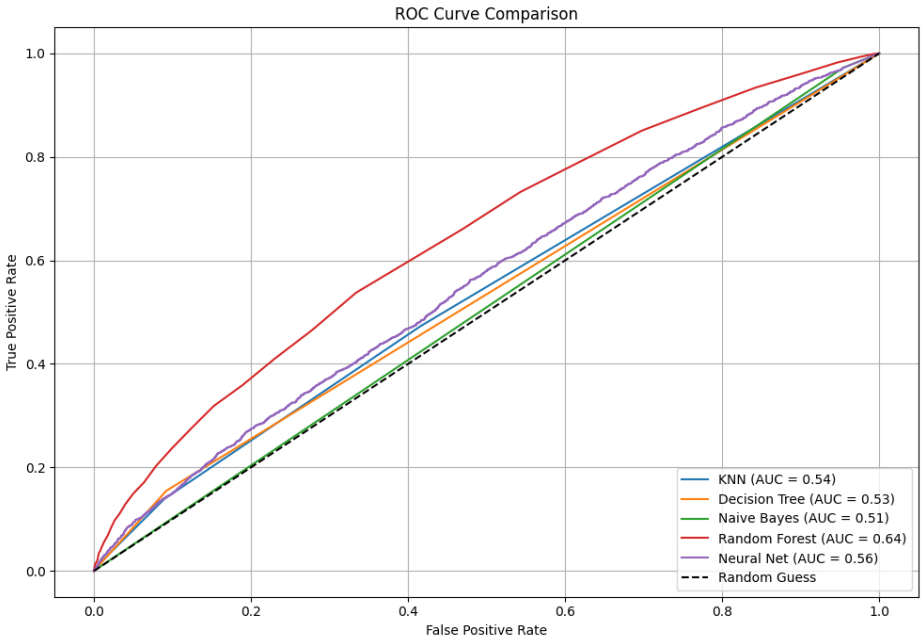
**Performance Comparison of ML Models**




*Note: Bar chart comparing Accuracy, Precision, Recall, and F1-Score for each model.*

**Figure 5**

**ROC Curve for All Classifiers**




*Note: Overlay ROC curves with AUC values for KNN, Decision Tree, Random Forest, etc.*

**Discussion**

The findings of this research reveal that the ensemble learning techniques like Random Forest are a suitable option to forecast readmission on imbalanced health care data. Their high generalizability and weighted F1-scores indicate that they generalize well. But the models always had a problem of forgetting positive cases, indicating that future research would benefit by dealing with overcoming this class imbalance using techniques such as Synthetic Minority Over-sampling Technique (SMOTE) or cost-sensitive learning. Feature selection and interpretability are also important areas of future research so that these models are not only accurate but also interpretable by clinicians. Despite that MLP showed excellent performance, its black-box nature and computationally complex algorithm limit

immediate usage in actual real-world clinical settings.

**Conclusion**

This paper bears witness to the applicability of hospital readmission forecasting of diabetic patients via machine learning algorithms. The optimal overall performance was achieved by the Random Forest model, which validates the ability of ensemble techniques in complex healthcare scenarios. However, none of the models' inability to identify the minority class of early readmissions precisely points towards the need for advanced resampling methods and additional tuning. Subsequent studies need to explore the real-time deployment possibility, deeper feature importance analysis, and hybrid modeling incorporating rule-based and data-driven methodologies to provide increased predictive accuracy and clinical assurance.

References

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